TCP-Relay (legacy, no longer maintained)

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	TITLE : TCP-Relay (legacy, no longer maintained)	

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1 Introduction

tcprelay is a TCP connection forwarder with load balancing capabilities. If compiled with TLS support, it may be used as SSL encryption wrapper.

1.1 Download

You can download the source code from the GitHub repository at https://github.com/MarcJHuber/event-driven-servers/. On-line documentation is available via https://projects.pro-bono-publico.de/event-driven-servers/doc/, too.

2 Operation

This section gives a brief and basic overview on how to run **tcprelay**.

In earlier versions, **tcprelay** wasn't a standalone program but had to be invoked by **spawnd**. This has changed, as **spawnd** is now part of the **tcprelay** binary. However, using a dedicated **spawnd** process is still possible and, more importantly, the **spawnd** configuration options and documentation remain valid.

tcprelay may use auxilliary **MAVIS** backend modules for authentication and authorization.

2.1 Command line syntax

The only mandatory argument is the path to the configuration file:

```
tcprelay [ -P ] [ -d level ]  [ -i child_id ] configuration-file [ id ]
```

If the program was compiled with CURL support, *configuration-file* may be an URL.

Keep the `-P` option in mind - it is imperative that the configuration file supplied is syntactically correct, as the daemon won't start if there are any parsing errors at start-up.

The `-d` switch enables debugging. You most likely don't want to use this. Read the source if you need to.

The `-i` option is only honoured if the build-in **spawnd** functionality is used. In that case, it selects the configuration ID for **tcprelay**, while the optional last argument *id* sets the ID of the **spawnd** configuration section.

2.2 Signals

Both the master (that's the process running the **spawnd** code) and the child processes (running the **tcprelay** code) intercept the `SIGHUP` signal:

- The master process will restart upon reception of `SIGHUP`, re-reading the configuration file. The child processes will recognize that the master process is no longer available. It will continue to serve the existing connections and terminate when idle.
- If `SIGHUP` is sent to a child process it will stop accepting new connections from its master process. It will continue to serve the existing connections and terminate when idle.

2.3 Event mechanism selection

Several level-triggered event mechanisms are supported. By default, the one best suited for your operating system will be used. However, you may use the environment variable `IO_POLL_MECHANISM` to select a specific one.

The following event mechanisms are supported (in order of preference):

- port (Sun Solaris 10 and higher only, `IO_POLL_MECHANISM=32`)

- kqueue (*BSD and Darwin only, `IO_POLL_MECHANISM=1`)
- /dev/poll (Sun Solaris only, `IO_POLL_MECHANISM=2`)
- epoll (Linux only, `IO_POLL_MECHANISM=4`)
- poll (`IO_POLL_MECHANISM=8`)
- select (`IO_POLL_MECHANISM=16`)

Environment variables can be set in the configuration file at top-level:

```
setenv IO_POLL_MECHANISM = 4
```

2.4 Configuration Syntax

A single configuration file is sufficient for configuring both **spawnd** and **tcprelay**. The basic format for this file is:

```
id = spawnd {
    # spawnd configuration directives
}

id = tcprelay {
    # tcprelay configuration directives
}
```

For example, the `spawnd` section could look similar to:

```
    listen = { port = 80}
    spawn = { exec /usr/local/libexec/tcprelay }
```

This tells spawnd to accept connections on the port given, and feed them to a **tcprelay** process. Please see the **spawnd** documentation for more configuration details.

tcprelay has its own set of configuration directives:

- `local address` = *addr*

  Specifies the local address used for outgoing connections.

- `rebalance` = *n*

  Re-balances peers after *n* requests. May be used to reactivate dead peers. Use with care. Default: unset.

- `remote = { ... }`

  The `remote` sections tell **tcprelay** where to relay connections to. Valid configuration directives inside the curly brackets are:

  - `address` = *IPAddress*
  - `port` = *TCPPort*
  - `protocol` = (`TCP` | `SCTP`)
  - `weight` = *Weight*

  Both the `address` and `port` directives are mandatory. The load balancing factor *weight* is optional and defaults to 1. Its value should somehow correspond to the load a destination can handle.

- `retire` = *count*

  If set, the daemon will terminate after processing *count* sessions, what may be useful to remedy the effects of memory leaks. By default, this is not set.

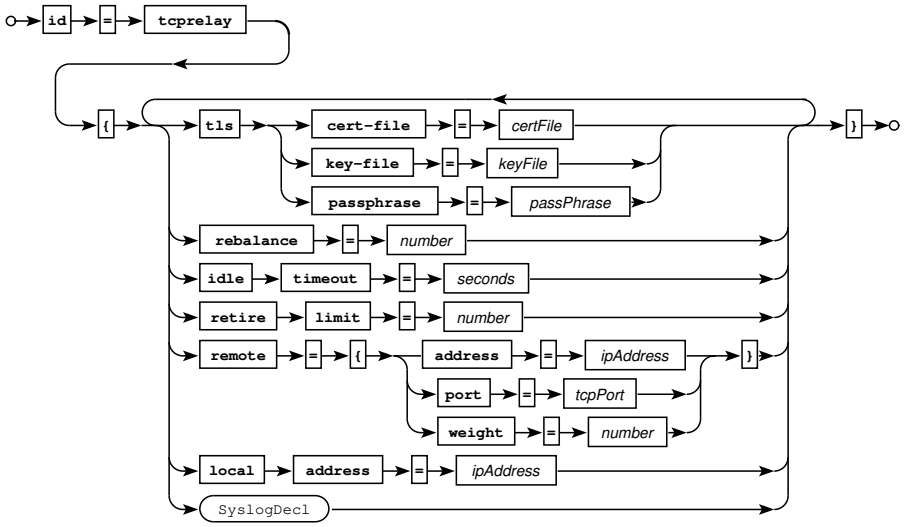
- `syslog` ((`ident` = Ident) | (`severity` = Level) | (`facility` = Facility))

  Selects syslog *ident*, *severity* and *facility*. Defaults to:

```
syslog ident = program-name
syslog facility = UUCP
syslog severity = INFO
```

- `idle timeout =`*Seconds*
  Set session timeout (default: `0`).
- `tls cert-file =` *CertFile*
  `tls key-file =` *KeyFile*
  `tls passphrase =` *PassPhrase*
  If compiled with TLS/SSL, *PassPhrase*, *CertFile* and *KeyFile* may be specified using this option.
  KeyFile may be omitted, it defaults to *CertFile*.

2.4.1 Railroad Diagram



Railroad diagram: TcprelayConfig

2.5 Sample configuration

```
#!/usr/local/sbin/spawnd
id = spawnd {
    listen = { port = 2222 }
    listen = { address = ::0 port = 2222 }
    listen = { ::0 port = 2224 }
    listen = { port = 2225 tls = yes }
    spawn {
        users max = 4000
        users min = 10
        servers min = 1
        servers max = 20
    }
}

id = tcprelay {
  remote = { address = 169.254.1.2 port = 22 }
  ssl cert = /some/where/sample.pem
  ssl passphrase =  12345
}
```

3 Bugs

- TLS re-negotiation is currently untested and may or may not work.

4 Copyrights and Acknowledgements

Please see the source for copyright and licensing information of individual files.

- **The following applies if the software was compiled with TLS support:**

  This product includes software developed by the OpenSSL Project for use in the OpenSSL Toolkit (http://www.openssl.org/).

  This product includes cryptographic software written by Eric Young (eay@cryptsoft.com").

- **Portions of the parsing code are taken from Cisco's tac_plus developers kit which is distributed under the following license:**

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